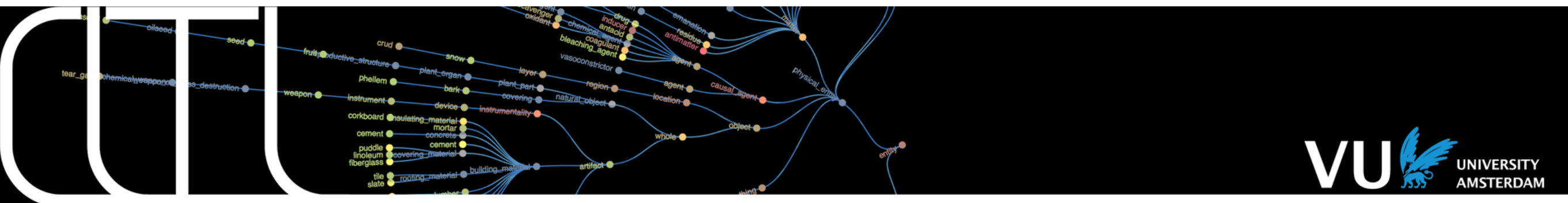


Text Mining 2026



Lecture 3: Machine Learning for NLP

Ilia Markov



Groups

- Lara Olders l.a.a.olders@student.vu.nl: Groups 1–10
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- Mateusz Kielan m.kielan@student.vu.nl: Groups 21–29
- Evelina Lune e.lune@student.vu.nl: Groups 30–38, 59
- Lazar Gyoshev l.gyoshev@student.vu.nl: Groups 39–47, 58
- Kevyn Smit k.m.smit2@student.vu.nl: Groups 48–57

Fridays **13:30-15:15**. Location: **NU-3A06** (and NU-3B05)

Preparation: <https://github.com/cltl/ba-text-mining>

Spreadsheets for questions

Overview

- Part I: NLP Pipelines
- Part II: Bag-of-Words classification
- Part III: Evaluation
- Part IV: Sequence classification
- Part V: Word Embeddings
- Part VI: Contextual Embeddings

Literature

- **Week 2:**

- Maynard et al. (2016), **Natural Language Processing for the Semantic Web: Chapter 2** (read sections on high-level explanations of NLP, can skip parts about particular software components - 2016)
- **Natural Language Processing with Python: Chapter 8**, Analyzing sentence structure, sections 1-5 (also see lab sessions)
- **Natural Language Processing with Python: Chapter 6**, Learning to classify (lab sessions)
- Dan Jurafsky and James H. Martin, **Speech and Language Processing: Chapter 4**, Logistic Regression and Text Classification (deeper explanation of Computational Linguistics: concepts and math behind)

- **Week 3:**

- Dan Jurafsky and James H. Martin, **Speech and Language Processing: Chapter 5** Embeddings (basis for deep learning)
- Wolf, T. et al., "**Transformers: State-of-the-art natural language processing.**" In Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations, pp. 38-45. 2020 (transformers in NLP)
- Church, K., Chen, Z., & Ma, Y. (2021). **Emerging trends: A gentle introduction to fine-tuning.** Natural Language Engineering, 27(6), 763-778. doi:10.1017/S1351324921000322 (how transformers are used to build NLP systems: SA, NEs, topics)

Part I & II

NLP Pipelines

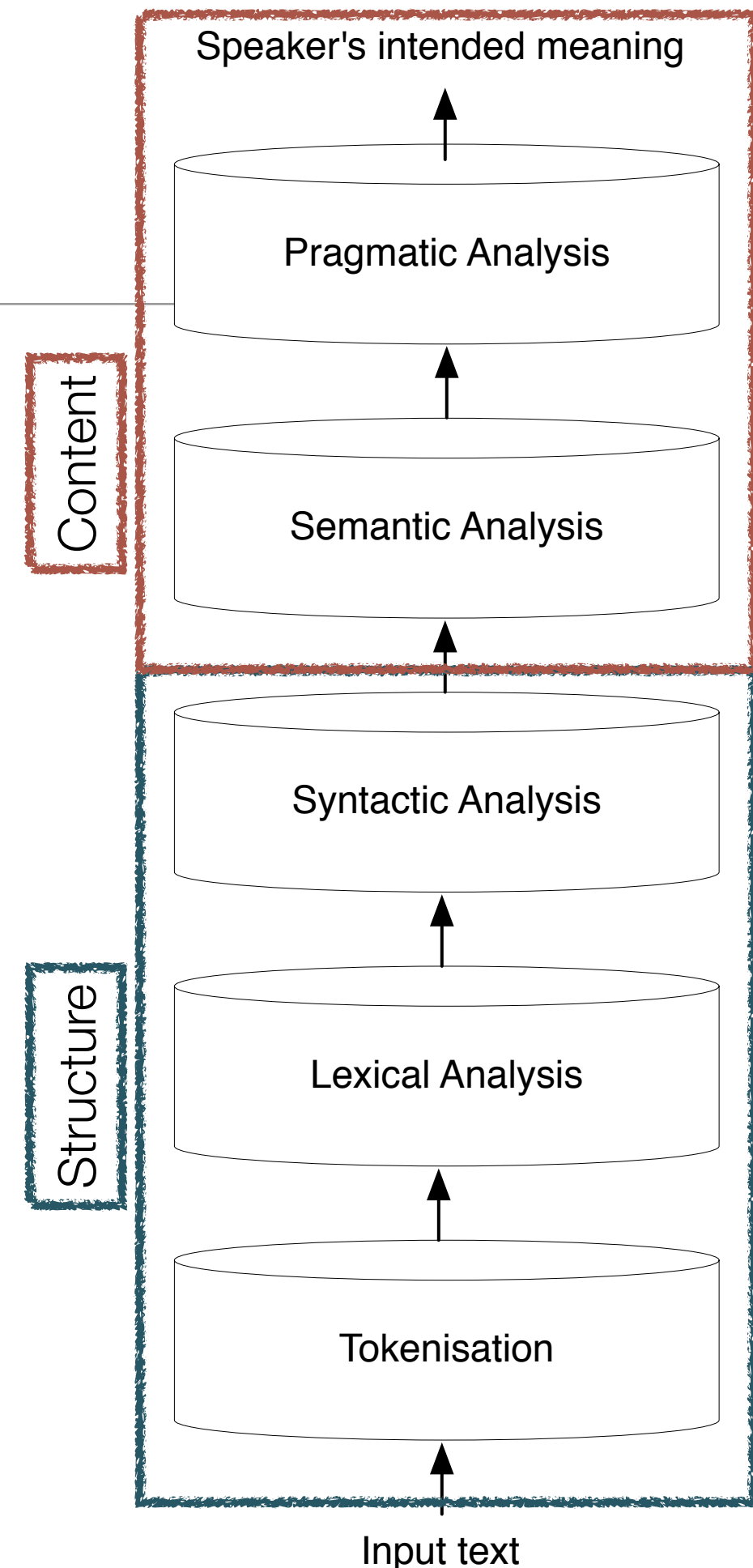
The art of digging text

Pipelines
Patterns and rules
Machine learning



Natural Language Processing

- Complex problem (extracting info from texts) is broken down into a number of smaller problems
- Simple, structural problems solved first and higher-level semantic tasks are solved later, using the output of earlier modules as input:
 - so-called pipeline architecture with dependencies across modules
 - error propagation
- For each problem different techniques:
 - knowledge-base & rules (linguistic knowledge)
 - machine learning (supervised and unsupervised), data driven



We always need to do preprocessing

- Even for the current state-of-the-art deep learning systems
- First problem, what is a word, what is a sentence? Not trivial!
- **Tokenization** (examples of problems)
 - nitty-gritty, data-base, (semi-)irony, \$523,45, 21st century, 9-11, Encoding issues (Latin, UTF-8/16, diacritics, “” quotes...), don't, men's, end-of-sentence hy-p-h-ens
- **Sentence splitting** (examples of problems)
 - Dr., Mrs., Bol.com, 7.5, etc. etc, white spaces, tables, formulae, titles, TABs, new lines, HTML markup <body><h1></h1><p></body>

All texts need preprocessing

- **Example from a news article:**

Documents filed to the San Jose federal court in California on November 23 list six Samsung products running the "Jelly Bean" and "Ice Cream Sandwich" operating systems, which Apple claims infringe its patents.

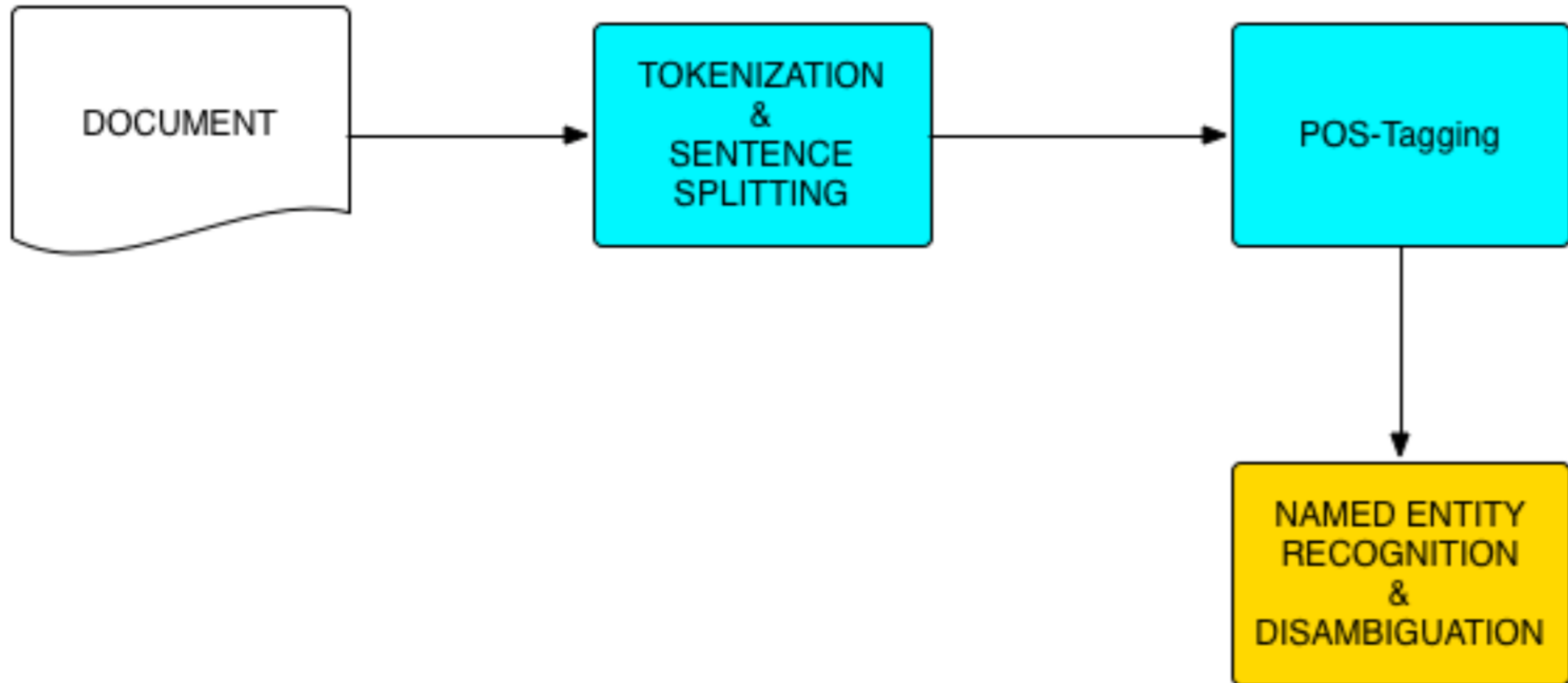
All text needs preprocessing

- **Example:**

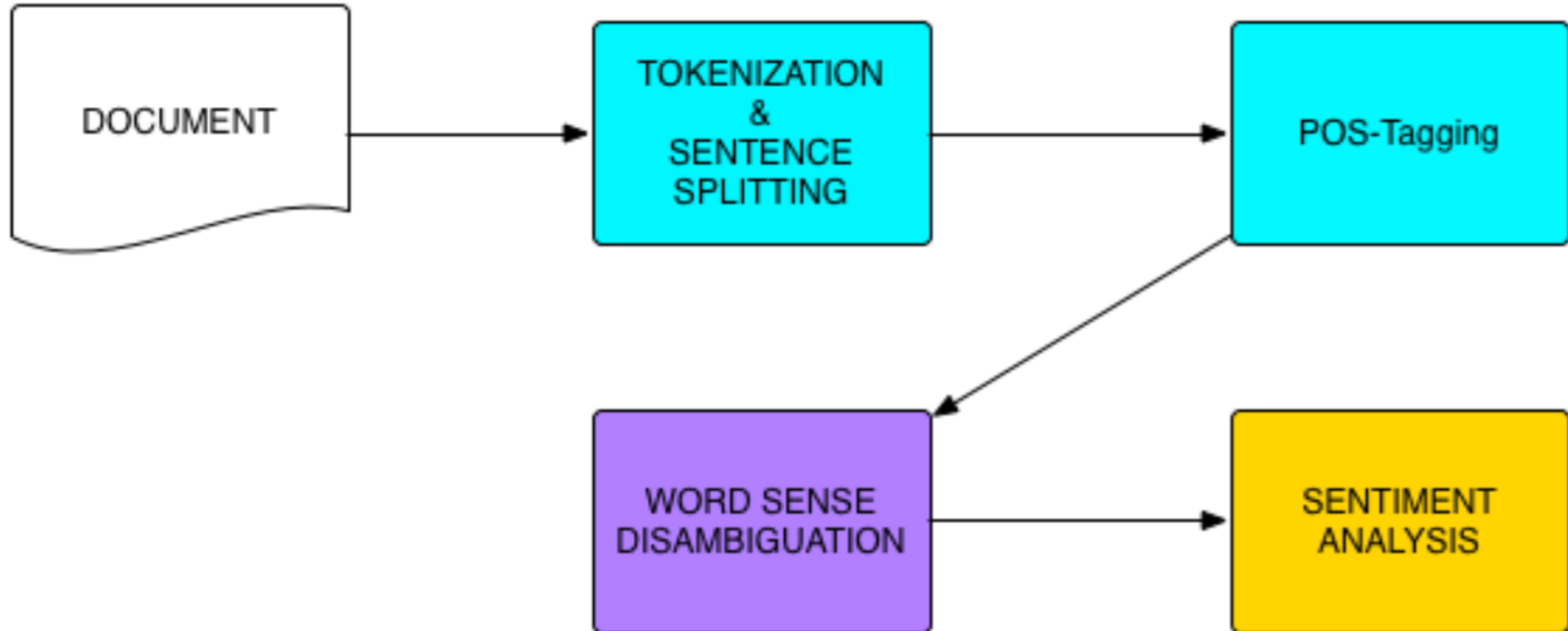
Documents filed to the San Jose federal court in California on November 23 list six Samsung products running the "Jelly Bean" and "Ice Cream Sandwich" operating systems, which Apple claims infringe its patents.

- *A single sentence ✓*
- *Documents —> documents —> document —> noun or verb?; list —> noun or verb? document = subject or object of list? (all affect grammatical structure)*
- *the "Jelly Bean" and "Ice Cream Sandwich" operating systems,*
 - *without punctuation: 9 tokens*
 - *including punctuation: if double quotes 14, if not 18 tokens (different number of tokens)*

Named Entity Recognition Pipeline Example



Sentiment Analysis Pipeline Example



Pipelines are likely to share the basic processing. Specific modules for higher-level interpretations.

Medical NLP Pipeline

scientific articles & clinical documents: admission notes, discharge summaries, radiology reports, pathology reports, etc.

X/HTML
to Text



PDF to
Text

tables,
references

Section
detection

abstract,
method,
related
work,
discussion

Sentence
Token

periods,
punctuation
(periods,
commas),
real
numbers,
initial case

Part-
of-Speech
Lemma

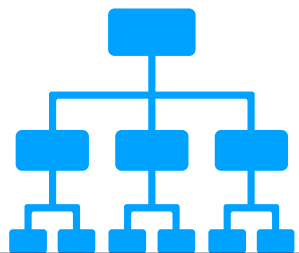
noun,
verb,
adjective
singular,
plural

Syntax

subject,
verb,
object

Named
Entity

gene,
protein,
drug,
disease



Structured Data

Hedging

unexpected
possible
unclear
15%
mortality

Time

March,
2005
last week
5 minutes

Relation

gene A
inhibits
gene B

Event

morbidities,
obesity,
smoking status,
treatment,
complications

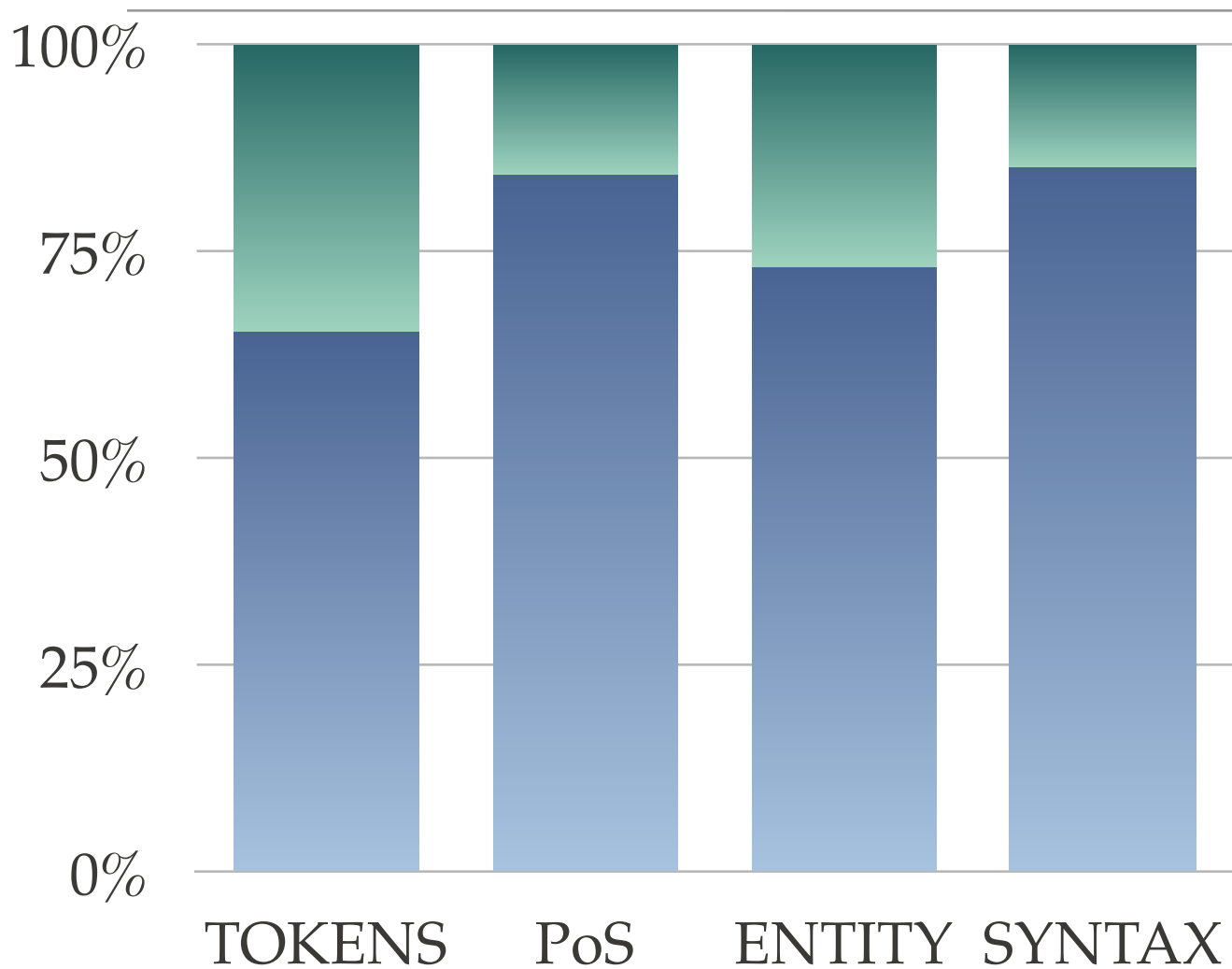
Entity
Linking

controlled
vocabulary,
medical
thesaurus,
ontology

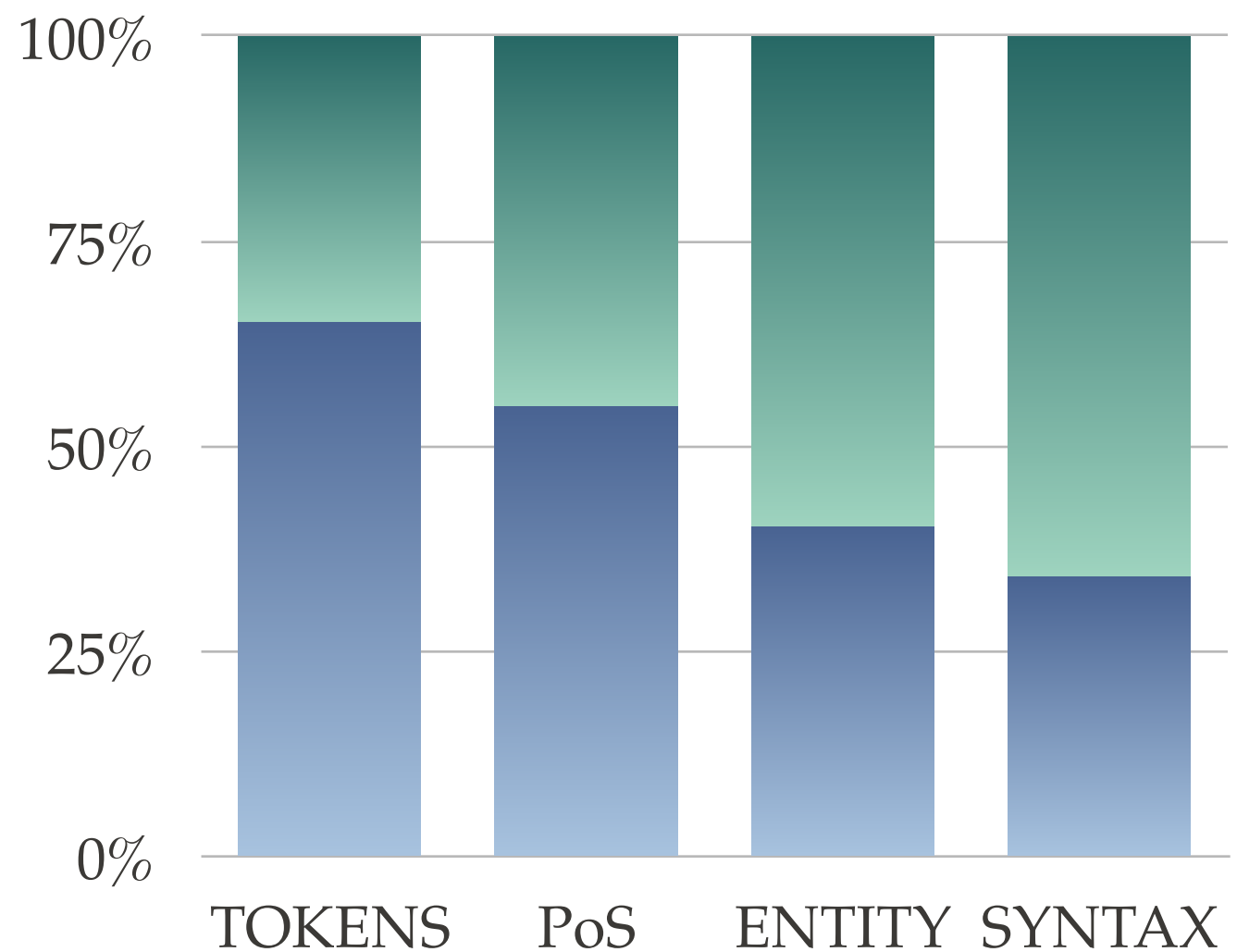
Some issues

- Dependencies across modules result in error propagation
- Ambiguities (multiple values with confidence score, e.g., POS tagging: 80% noun, 20% verb) are often not exploited by next levels
- Conflicts: different modules state information that is not compatible
- Complex and difficult to maintain, e.g., input and output needs to be interoperable across modules

Error propagation



Isolated results for
cross-folded
training benchmarks



Propagated results for
cross-folded
training benchmarks

Performance in relation to domain specificity and complexity

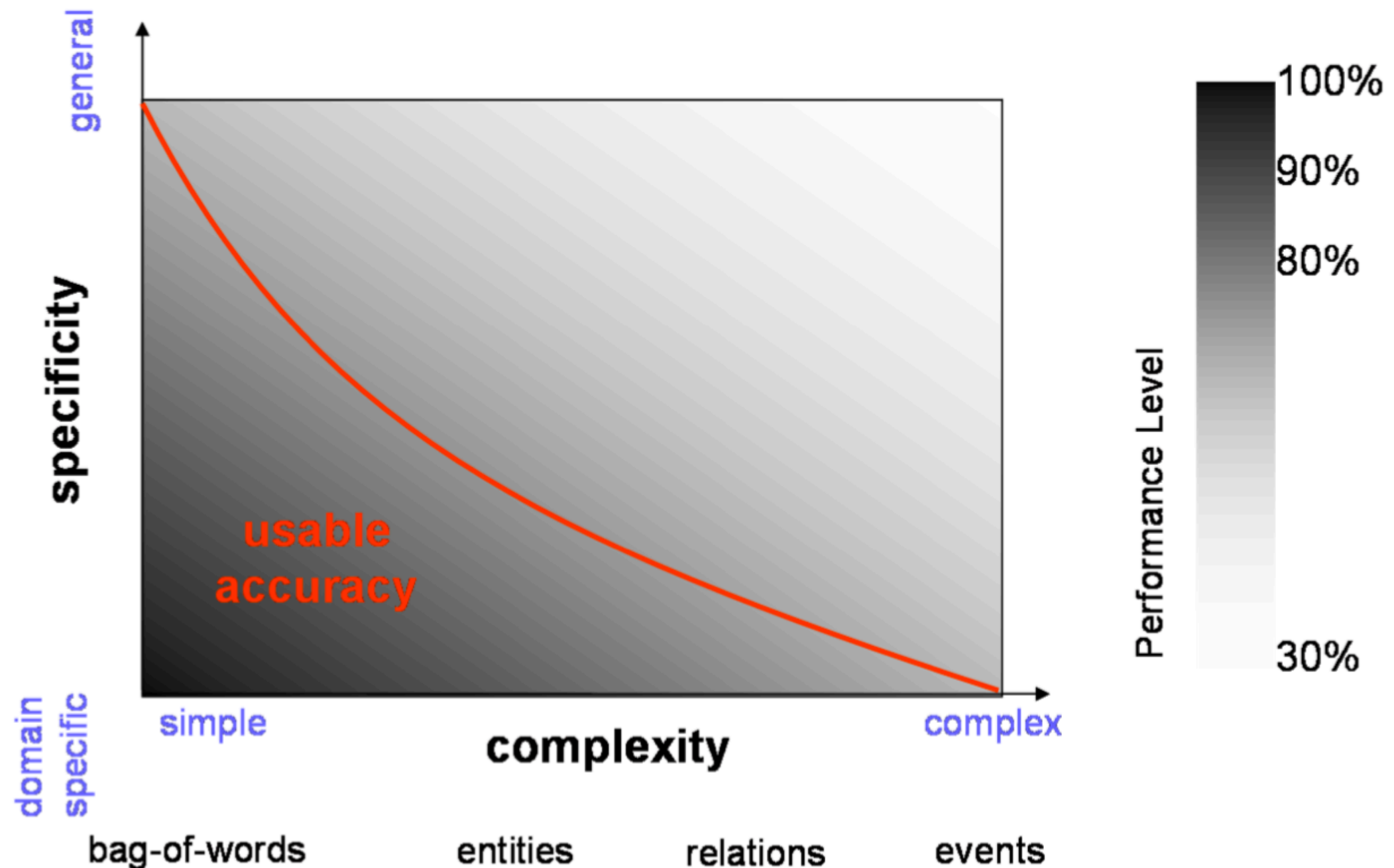


Figure 1.3: Performance Tradeoffs for NLP tasks

Two main problems for Text Mining

Ambiguity & Variation

Ambiguity

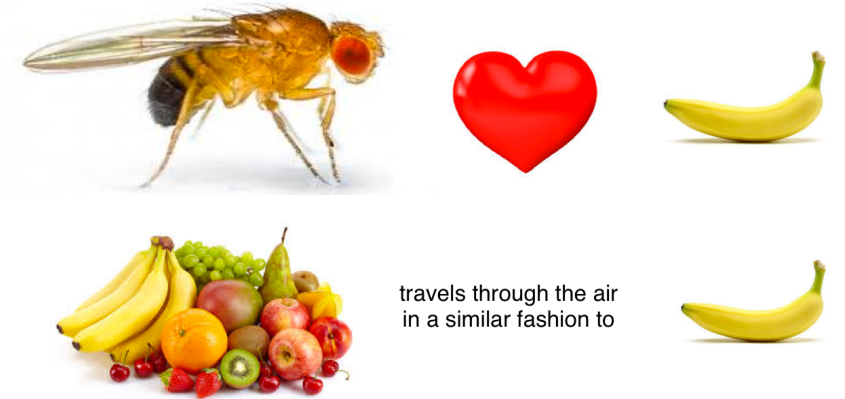
How many meanings for a word?

How many relations in a sentence?

Ambiguity

- *Structural ambiguity:*

- “*Fruit flies like a banana*”



- *Lexical ambiguity: words have more than one meaning (polysemy)*

News headlines

- Foot Heads Arms Body.
- Hospitals Sued by 7 Foot Doctors.
- British Left Waffles on Falkland Islands.
- Stolen Painting Found by Tree.

Lexical ambiguity is pervasive and we do not perceive it!!!

121 most frequent English nouns have on average 7.8 meanings each and account for about 20% of word occurrences in real text (in the Princeton WordNet (Miller 1990), according to Ng and Lee (1996)).

“He gave a soft ball across the line from the center of the field, making a major point and giving a minor lead.”

The meaning puzzle

Ambiguity demo

Sentence:

He(2) gave(44) a soft(19) ball(12) across the line(30) from the center(18) of the field(17) , making(49) a maj

Compute

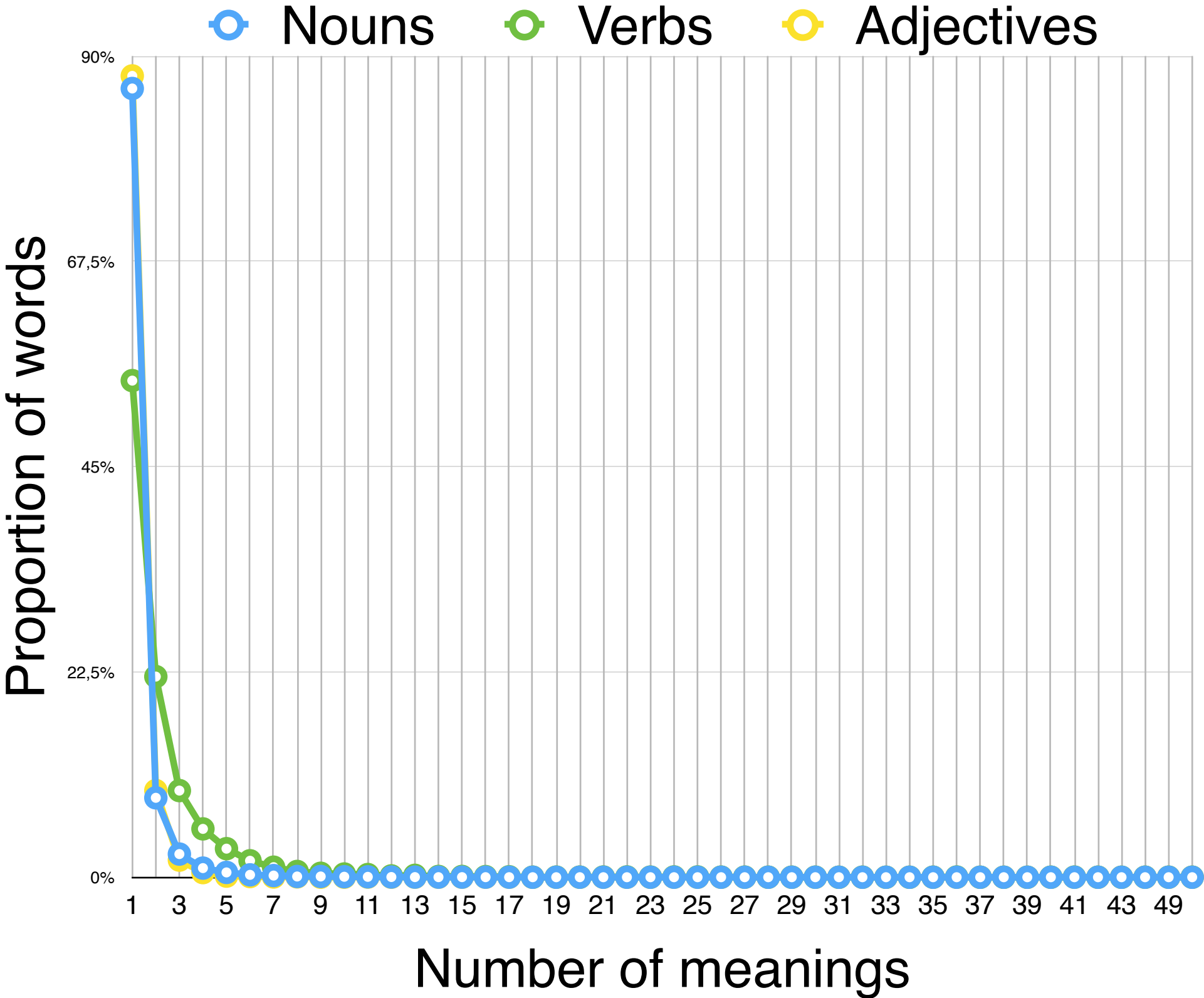
He(2) gave(44) a soft(19) ball(12) across the line(30) from the center(18) of the field(17) ,
making(49) a major(8) point(26) and giving(44) a minor(10) lead(17) . = 14041749244723200
possible meaning combinations

14,041,749,244,723,200

Number of senses for the words are taken from WordNet 3.0 (Fellbaum, 1998)

Puzzle for a computer

Ambiguity in WordNet3.0



Pol.	Nouns	Verbs	Adj.
1	101863	6277	7313
2	10257	2536	791
3	2989	1095	156
4	1178	611	42
5	620	361	10
6	306	208	6
7	212	127	2
8	94	73	3
9	96	50	
10	60	41	
11	48	34	1
12	25	17	
13	14	24	
14	8	10	
15	10	10	
16	6	8	
17	5	6	
18	1	1	
59		1	
	117792	11490	8324

[https://en.wikipedia.org/wiki/Head_\(disambiguation\)](https://en.wikipedia.org/wiki/Head_(disambiguation))

Most
ambiguous
words in
WordNet3.0
are frequent

Nouns	Polysemy	Verbs	Polysemy	Adjectives	Polysemy
head	33	break	59	active	11
line	30	make	49	soft	8
point	26	give	44	light	8
base	20	take	42	inactive	8
case	20	cut	41	major	7
cut	20	run	41	minor	7
center	18	carry	40	open	6
field	17	draw	36	dry	6
lead	17	get	36	direct	6
shot	17	hold	36	hard	6
stock	17	play	35	heavy	6
play	17	fall	32	short	6
position	16	go	30	long	5
break	16	catch	29	critical	5
run	16	call	28	cut	5
place	16	work	27	straight	5
pass	16	raise	27	immature	5
form	16	turn	26	uncut	5
Service	15	cover	26	offensive	5
mark	15	check	25	national	5
order	15	pass	25	right	5
light	15	charge	25	syllabic	5
bar	15	set	25	single	4

Variation

How many words to name/make reference to something?

How many expressions and stories for an event?

Variation in expressions

Different words and expressions

- BNC Holdings Inc named Ms G Torretta as its new chairman.
- Nicholas Andrews was succeeded by Gina Torretta as chairman of BNC Holdings Inc.
- Ms. Gina Torretta took the helm at BNC Holdings Inc.

Person can be referred differently. Many ways you can express this event.

Information spread over multiple sentences involving coreference

- After a long boardroom struggle, Mr Andrews stepped down as chairman of BNC Holdings Inc. He was succeeded by Ms Torretta.

5,000 words for person in WordNet

mortal; posturer; controller; **withholder**; suppressor; subduer; fugitive; divider; subdivider; outcaste; bereaved_person; yielder; nude; streaker; **unperson**; baby_buster; neighbor; loose_cannon; ladino; communicator; announcer; town_crier; caller; caller; muezzin; hisser; gossipmonger; yenta; cat; telltale; **scandalmonger**; allegoriser; presenter; promisee; quoter; transmitter; spammer; answerer; hedger; assenter; interviewee; **don'tknow**; testee; passer; popularizer; avower; laudator; clapper; waffler; wirer; conferrer; confessor; reporter; newswoman; television_reporter; anchorman; avower; postulator; author; gagwriter; poet; sonneteer; poetess; homer; elegist; frost; key; gilbert; gray; pound; poet_laureate; odist; spender; poet_laureate; bard; biographer; autobiographer; hagiographer; novelist; folk_writer; folk_poet; **cyberpunk**; west; wood; authoress; abstractor; pamphleteer; speechwriter; drafter; paragrapher; space_writer; tragedian; snow; day; playwright; rice; kid; cooper; buck; scriptwriter; film_writer; wordmonger; framer; literary_hack; rand; coauthor; scenarist; litterateur; **word-painter**; wordsmith; alliterator; sand; e._e._cummings; heller; grass; spark; journalist; photojournalist; reed; stone; sob_sister; sports_writer; newspaperwoman; war_correspondent; foreign_correspondent; broadcast_journalist; gazetteer; columnist; newspaper_columnist;

Many of these words are contextual, they refer to people in particular roles.

4,000 words for movements

go; swim; buoy; drive; island_hop; whistle; ski; slalom; hot-dog; wedel; water_ski; schuss; pass_over; breeze; err; return; revisit; retrace; cut_back; resurrect; return; home; head_home; double_back; bounce; boomerang; fly; come; retrograde; walk; constitutionalize; speed; bang; **swash**; tread; step_on; beetle; circulate; drift; swim; bucket_along; shoot; rip; barge; dash; plunge; hurtle; sit; canter; override; prance; ride_herd; ride_horseback; prance; post; trot; gallop; canter; gallop; outride; lance; scramble; plough; sift; **zigzag**; billow; pursue; haunt; tail; tree; hound; ferret; run_down; quest; stalk; roll; troll; bowl; travel_purposefully; wend; whisk; cruise; stooge; steamer; go_forward; head; make; trace; limp; wander; roar; ease; circulate; float; ride; shack; draw; caravan; career; raft; swap; thrash; retreat; cocoon; automobile; step; backpedal; blow; stream; tide; waft; crawl; formicate; slice_into; run; precede; lead; draw_away; travel_along; ascend; heel; turn; angle; push; travel; ride; fly; cruise; ship; sail; wind; snake; pan; repair; taxi; precess; cast; **jazz_around**; maunder; travel; iterate; go_up; rise_up; resurface; well; intumesce; emerge; uprise; ferry; transfer; betake_oneself; march_on; string; edge; forge; penetrate; ratchet_up; sneak_up; plough_on; draw_in; slide_by; fell; impinge; overtake; clear; hop; get_by; tram; prance; derail; go_through; get_across; stride; take; ford; tramp; jaywalk; crisscross; bridge; walk; hop; course; cut; muscle; lock; negotiate; pass_through; make; jostle; bushwhack; claw; pass_over; cut; crash; transit; blunder; cycle; cycle_on; squeak_by; break_through; run; overstep; go_around; drive; pull; cut_in; wing; fly_on; soar; rack; buzz; hover; poise; flight; go_down; go_down; drip; correct; subside; dismount; pitch; go_down; founder; subside; submerge; dive; belly-flop; jackknife; flop; rope_down; cascade; drop; flump_down; decline; dip;

2,037 words for noise

grinding; racket; report; squeak; clap; clack; snore; chatter; chattering; brouhaha; hubbub; uproar; **katzenjammer**; clatter; shrieking; scream; screech; screaming; shriek; screeching; blaring; blare; din; cacophony; clamor; grumble; rumble; grumbling; rumbling; squawk; plump; crackling; crackle; crepitation; decrepitation; snap; explosion; squish; rhonchus; hum; humming; **swoosh**; **whoosh**; clangoring; clang; clank; clash; clangor; crash; clangour; whisper; whispering; rustling; rustle; chug; sizzle; plonk; howl; **squeal**; plop; scrape; scratching; scratch; scraping; chatter; chattering; ding-dong

This is how much variation you can expect when people refer or describe something in relation to how much ambiguity there is.

Natural language generation

- Finding the correct expression for data, images, signals
- Applications: Machine translation, Text summarisation, Chatbots and Dialogue systems, Robo-journalism (computer generates a news article)
- Most systems stay on the safe side by generating basic or generic expressions with little information:

- Image labeling: *person, (young) man, boy* rather than *hooligan, protester, criminal* (informative but risky, especially with respect to biases in the data)
- Chatbots similar problem (very general answer):

- Human: *I like tennis,*
- Machine: *tell me more* rather than *Are you a fan of Federer or Nadal?*



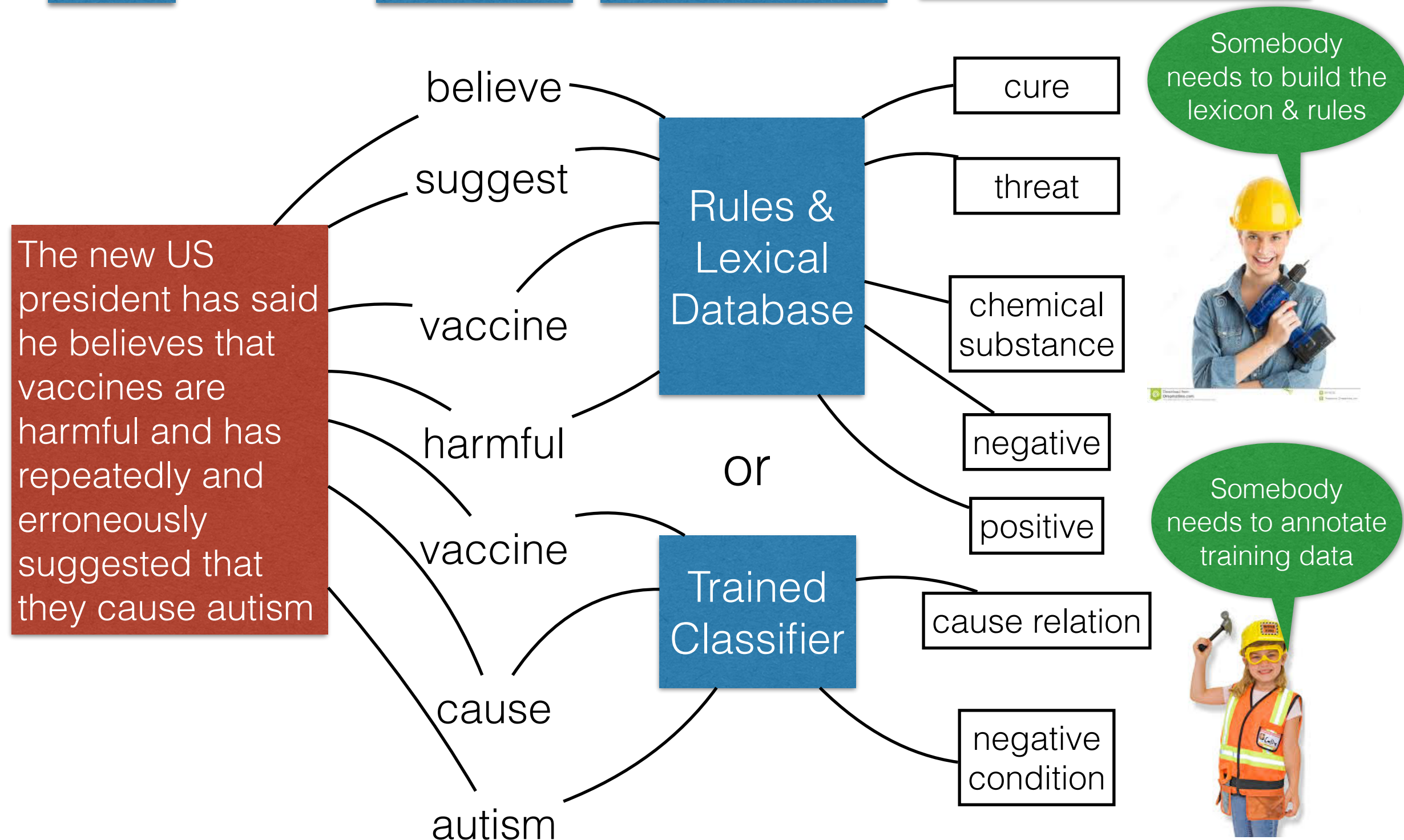
Natural language understanding

Text

Words

Resource

Interpretation



NLP Approaches

- **Rule-based:** tell the machine exactly what to do under specific circumstances (become too complex, still used by companies)
- **Machine learning:** the machine learns to associate patterns with interpretations (outperform rule-based systems, difficult to control: biases, other side effects)
 - Supervised machine learning: *the machine uses labelled examples to learn*
 - Unsupervised machine learning: *the machine identifies patterns without using labelled examples (learns about the language in general by processing a massive amount of data)*
- **Hybrid** = a combination of (un)supervised machine learning and rule-based approaches

Rules and lexicons!

● **lexicon** that lists words with properties, e.g. sentiment scores for positive (good, excellent) or negative sentiment (bad, worthless):

- **Ambiguity:** *low* prices $\longleftrightarrow$ *low* ceilings
- **Other issues:**
 - **negation** “staff is not friendly but the place is clean”
 - **intensifiers** “very friendly”

● **Examples of lexical and rule-based analysis:**

- Sentiment: VADER (for tweets); <https://github.com/cjhutto/vaderSentiment>
- Emotions: NRC (emotions expressed by words): <https://saifmohammad.com/WebPages/NRC-Emotion-Lexicon.htm>
- Psychological features: LIWC <http://liwc.wpengine.com>

VADER lexicon

Valence Aware Dictionary and sEntiment Reasoner

TOKEN, MEAN-SENTIMENT-RATING, STANDARD DEVIATION, and RAW-HUMAN-SENTIMENT-RATINGS [10 raters -4, 0, +4]

```
arrogance    -2.4    0.66332 [-3, -2, -2, -3, -1, -2, -3, -3, -2, -3]
arrogances   -1.9    0.53852 [-1, -2, -3, -2, -2, -2, -2, -1, -2, -2]
arrogant     -2.2    0.6    [-2, -2, -2, -3, -2, -3, -3, -2, -1, -2]
arrogantly   -1.8    1.4    [-3, -2, 2, -3, -1, -3, -2, -2, -2, -2]
ashamed      -2.1    1.3    [-3, -3, -3, -2, -2, 1, -2, -4, -1, -2]
ashamedly    -1.7    0.64031 [-2, -2, -2, -1, -2, -3, -1, -2, -1, -1]
ass          -2.5    1.43178 [-4, -1, -2, -1, -3, 0, -2, -4, -4, -4]
```

```
😊 grinning face
😄 beaming face with smiling eyes
😂 face with tears of joy
🤣 rolling on the floor laughing
😄 grinning face with big eyes
😄 grinning face with smiling eyes
😓 grinning face with sweat
😏 grinning squinting face
😉 winking face
```

NRC emotion lexicon annotated by the crowd

<https://saifmohammad.com/WebPages/NRC-Emotion-Lexicon.htm>

☉ beating anger 1

☉ beating anticipation 0

☉ beating disgust 0

☉ beating fear 1

☉ beating joy 0

☉ beating negative 1

☉ beating positive 0

☉ beating sadness 1

☉ beating surprise 0

☉ beating trust 0

☉ betrayal anger 1

☉ betrayal anticipation 0

☉ betrayal disgust 1

☉ betrayal fear 0

☉ betrayal joy 0

☉ betrayal negative 1

☉ betrayal positive 0

☉ betrayal sadness 1

☉ betrayal surprise 0

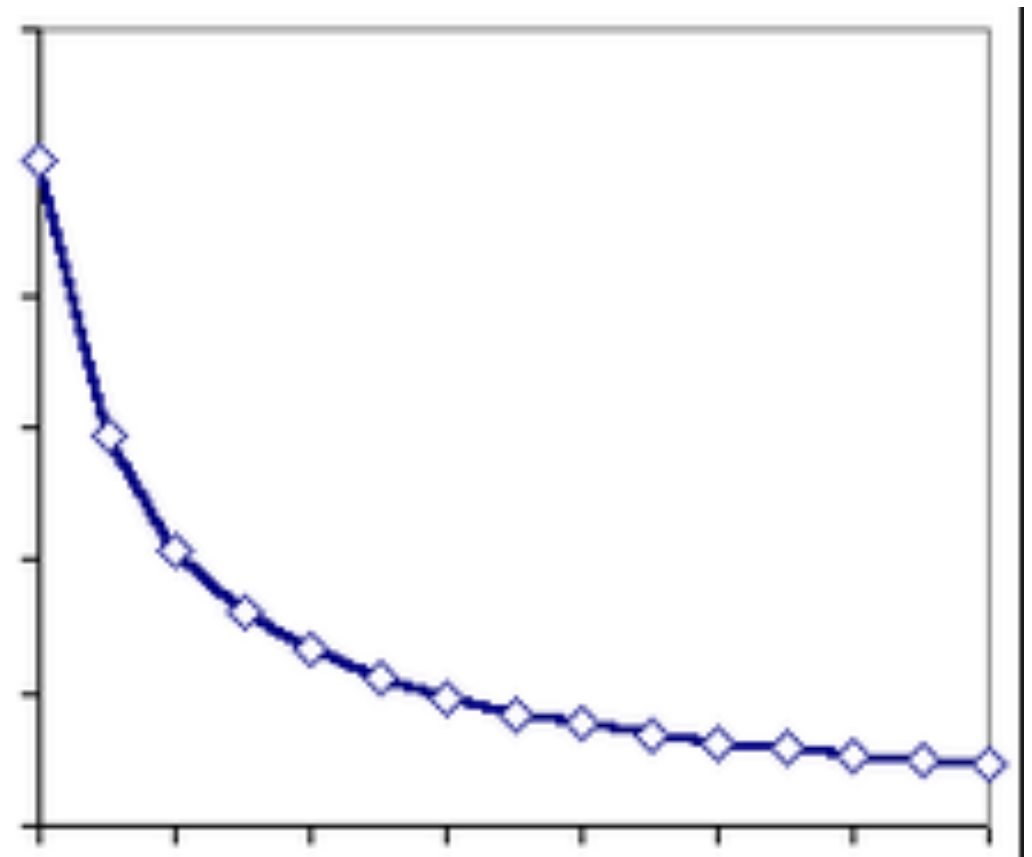
☉ betrayal trust 0

Formulating precise rules

● Easy to get some results quickly but impossible to cover all ways of expressing information

● **Law of diminishing returns:**

- $\text{Effort}(X)^N$
- $N = 1 \rightarrow 50\%$ recall
- $N = 2 \rightarrow 60\%$
- $N = 3 \rightarrow 65\%$
- $N = 4 \rightarrow 68\%$

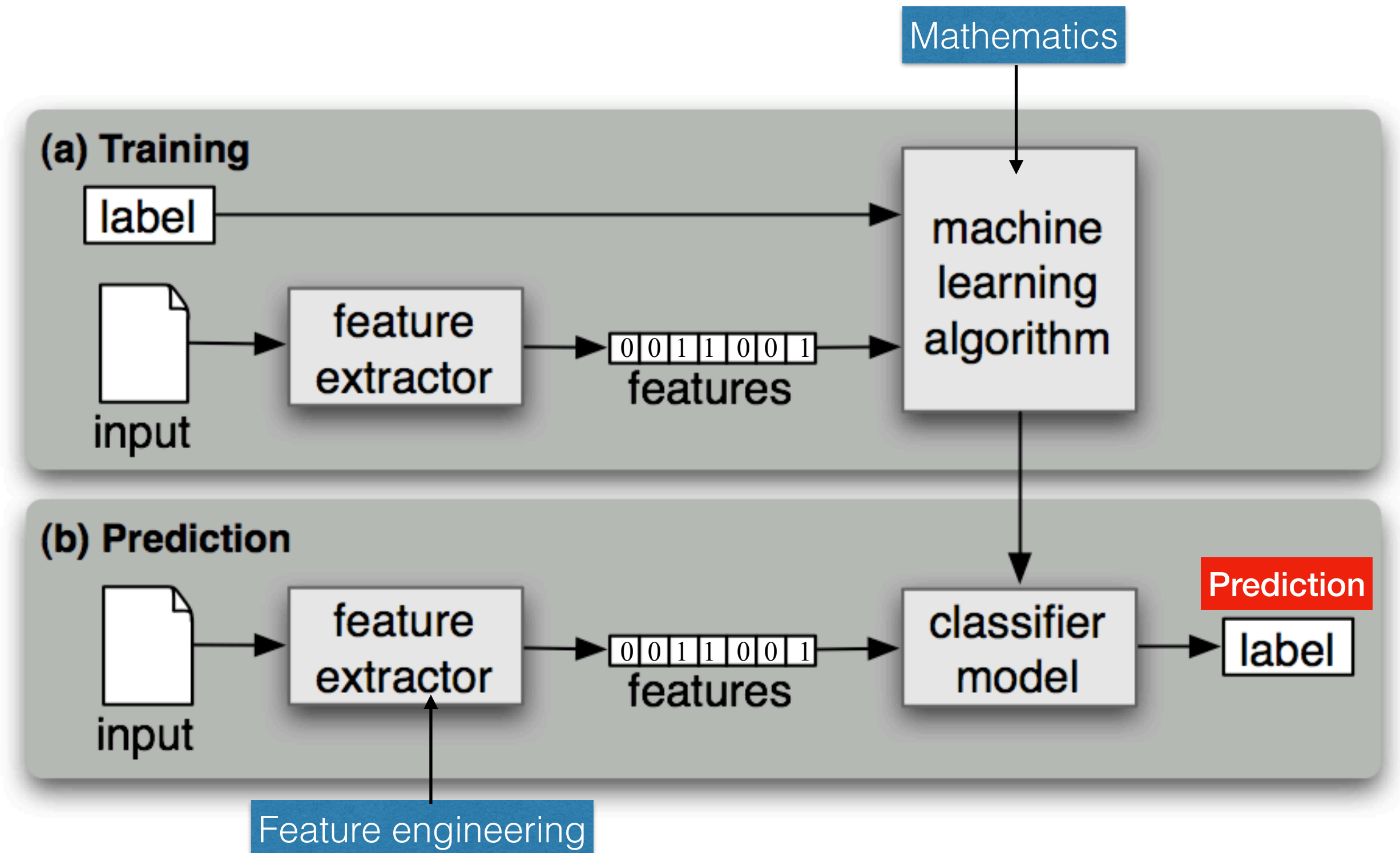


Increasing recall requires more and more effort:
adapting rules, lexicon, making it more complex

Why machine learning?

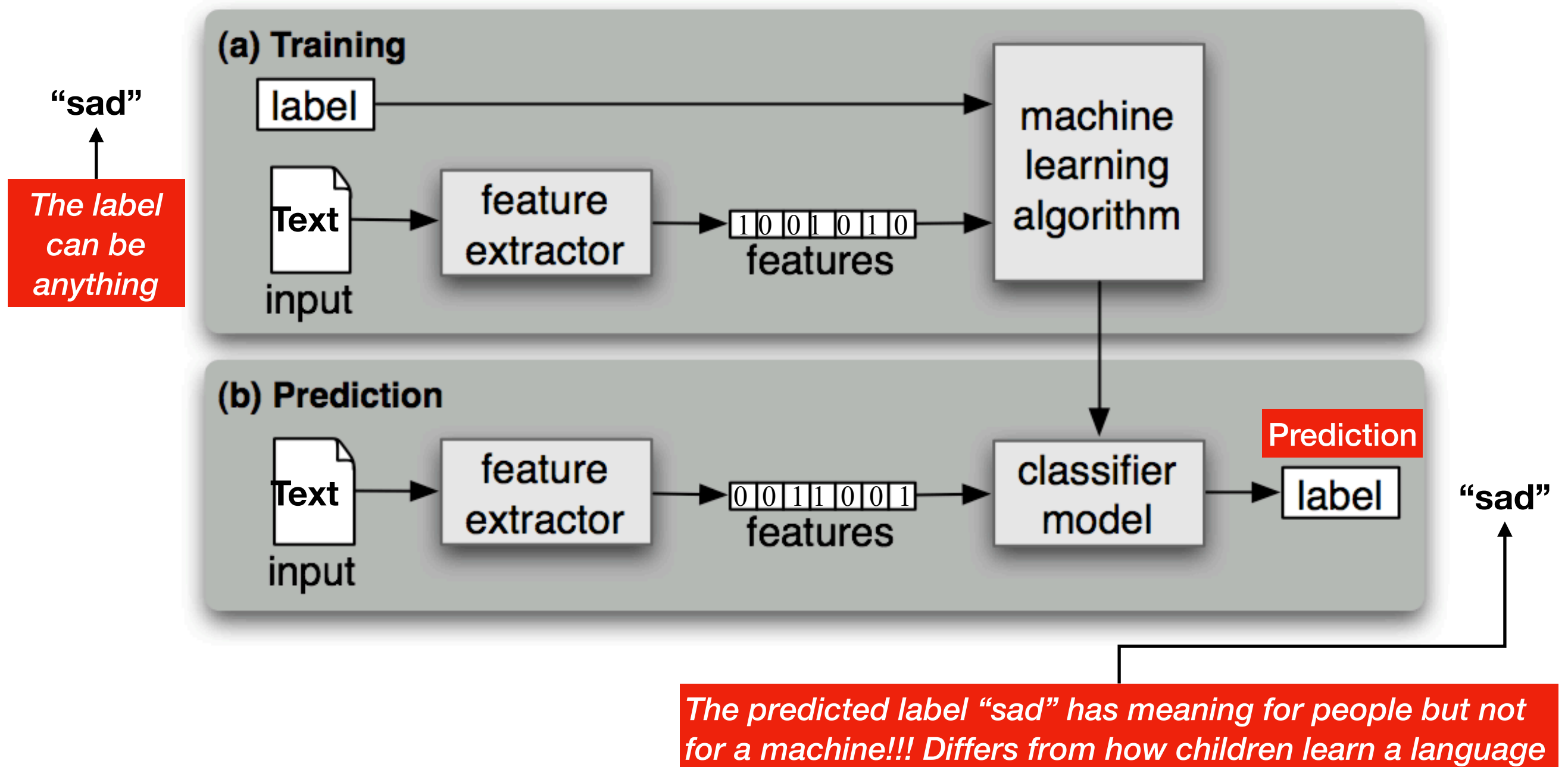
- Hand-crafted models (e.g., grammars) failed empirical tests
- Variation and dynamics of language is far bigger than realised
- Rule systems too complex:
 - difficult to maintain
 - psychologically unrealistic (do children learn a language by learning these rules?)
- Machine learning appears to work better than any set of the rules we invented so far

Supervised Machine learning (NLTK book)



a lot of

Machine Learning needs data!



Supervised machine learning systems

- **Training corpus:** sample of **representative texts** “*annotated*” by people with interpretation **labels**, e.g., part-of-speech of words, the meaning of words, entity phrases, syntactic dependencies, events, sentiments, etc.
- **Tag set:** the **set of labels** described in a “code book” with examples and decision criteria: different for each task and at different (hierarchical) levels, e.g. topics and subtopics, fine-grained parts-of-speech tags, word meanings in a lexicon or Wikipedia. The more fine-grained the tag set is, the more complex it would be for people assign tags.
- **Feature selection:** **represent text as a set of features** such that a computer can compare the training texts $T_{1..t}$ **associated with label** $L_{1..t}$ with a **test text** T_{test}
- **Classifier:** “match” features from training texts $T_{1..t}$ to the features of the target text T_{test} to predict the most likely label L_{test}

NLP Datasets

- **Scientific competitions, shared tasks:** MUC (Message Understanding Conference), TREC (information retrieval), CLEF (cross-lingual retrieval), CoNLL (natural language learning), SensEval (word sense detection), SemEval (semantic tasks), PAN (digital text forensics and stylometry), many more....
- Machine learning community: **Kaggle**
 - <https://www.kaggle.com/ratatman/sentiment-lexicons-for-81-languages>
- Some **blogs**:
 - <http://hatespeechdata.com>
 - <https://github.com/niderhoff/nlp-datasets>
 - <https://blog.cambridgespark.com/50-free-machine-learning-datasets-natural-language-processing-d88fb9c5c8da>

New datasets are released constantly

Data formats

- Text, CSV, TSV, XML, JSON, HTML
- Inline annotations (inside the text)
- (Layered) Stand-off annotations (point to the text, not inside the text)
- DOC, PDF, Excel are usually converted to above

Representing document with sentiments

- Example of a dataset from NLTK (lab sessions)
- `nltk_data/corpora/movie_reviews`
 - subfolder “pos”: 1,000 reviews as text files
 - subfolder “neg”: 1,000 reviews as text files
- `from sklearn.datasets import load_files`
(there are softwares that expect this kind of data structure, and you can generate the training set with just one command)

WASSA-2017

Shared Task on Emotion Intensity

ID	Tweet	Label	Score
2171	Just had my interview for the medical internship and I am beyond nervous 😞 I was shaking so ba	fear	1,000
4102	I feel like I am drowning. #depression #anxiety #falure #worthless	sadness	1,000
3121	OMG NOL i love u so so much i love ur cheery face and hugs 🥰🥰🥰 @CRAZYmac17	joy	0,980
2133	like srsly somebody help me deal with this social anxiety	fear	0,979
2151	@Twitch how do I stop that horror movie themed commercial? Suddenly hearing screams is really	fear	0,979
3106	I'm so excited to see Nat tonight 😍😍.. And how happy and cheery she is! & then I'm even	joy	0,979
4110	i am le depressed i I hate myself xDDdD #sad #depress #loner #emo	sadness	0,979
1157	Who the hell is drilling outside my house?! Literally got to sleep at half four after a busy shift and t	anger	0,976
2148	Panic attacks are the worst. Feeling really sick and still shaking. I should be a sleep. #anxiety #de	fear	0,958
3156	@priny_baby happpppy happpppyyyyyy happpppppyyyyy haaapppyyyy birthday best friend!! Love y	joy	0,958
4116	I want to pour all my tears on someone right now. So tired of this #upset #sad	sadness	0,958
4132	Honestly don't know why I'm so unhappy most of the time. I just want it all to stop :(#itnevergoe	sadness	0,949
3141	Finally have all my braces on and I can't stop smiling 😊😊😊😊😊	joy	0,940
3155	Howling with laughter at "WELL DONE BEZZA!" #bakeoff #GBBO	joy	0,940
1098	Stop tracking back you fucking potato faced cunt errrr infuriating 😡😡😡😡😡 #angry #Rooney	anger	0,938
2126	Been up since 4am. Too scared to go back to sleep #nightmare — feeling scared	fear	0,938
2158	So nervous I could puke	fear	0,938
2199	@AnthonyEinzig I'm so sorry. This is heartbreaking and so scary. If it can happen there it can happ	fear	0,938

“inline” Annotation

- (4) a. Perhaps without realizing it, Mr. Taffner simultaneously has put his finger on the problem and an ideal solution: “Capital City” should have been a comedy, a worthy sequel to the screwball British “Carry On” movies of the 1960s.
- b. Perhaps without realizing it, Mr. <ENAMEX TYPE=“PERSON”>Taffner</ENAMEX> simultaneously has put his finger on the problem and an ideal solution: <ENAMEX TYPE=“WORK_OF_ART”>“Capital City”</ENAMEX> should have been a comedy, a worthy sequel to the screwball <ENAMEX TYPE=“NORP”>British</ENAMEX> “<ENAMEX TYPE=“WORK_OF_ART”>Carry On</ENAMEX>” movies of <ENAMEX TYPE=“DATE”>the 1960s</ENAMEX>.

```
( (S (NP-SBJ (NNP Bartok))
  (VP (VBZ describes)
      (NP (NP (DT the) (NN form))
        (PP (IN of)
          (NP (DT the) (JJ first) (NN movement))))))
  (PP-CLR (IN as)
    (NP (NP (ADJP (ADVP (" ")) (RBR more)
      (CC or) (RBR less)) (JJ regular))
      (NN sonata) (NN form))))
  (, .)
```

CONLL: Computational Natural Language Learning

<https://www.conll.org>

CONLL format to represent data: each token on a separate line followed by TAB separated columns with annotations (TSV)

text = The American Museum of
Natural History was established
in New York in 1869.

1	The	O
2	American	B-Organization
5	Museum	I-Organization
6	of	I-Organization
8	History	I-Organization
9	was	O
10	established	O
11	in	O
12	New	B-Location
13	York	I-Location
14	in	O
15	1869	O
16	.	O

IO(B) style
I = insight
O = outside
B = beginning

text = The government closed
the National Museum of War
after the protests.

1	The	?
2	government	?
5	closed	?
6	the	?
8	National	?
9	Museum	?
10	of	?
11	War	?
12	after	?
13	the	?
14	protests	?
15	.	?

Learn patterns for sequences of tokens

Apply patterns to sequences of tokens

<https://www.clips.uantwerpen.be/conll2003/ner/>

Layered stand-off Annotation

- Raw text is not modified
- Offset pointers to the raw text layer
- Annotation layers with identifiers (lemma, POS, etc.)
- Layers can point to each other or to offsets
- New layers can be added easily
- Alternatives do not complicate the representation

Newsreader Annotation Format (NAF)

```
<?xml version="1.0" encoding="UTF-8"?>
<NAF xml:lang="en" version="v3">
  <nafHeader> ☐☐☐ </nafHeader>
  <raw><![CDATA[Qatar Holding sells 10% stake in Porsche to founding families.]]></raw>
  <text>
    <wf id="w1" sent="1" para="1" offset="0" length="5">Qatar</wf>
    <wf id="w2" sent="1" para="1" offset="6" length="7">Holding</wf>
    <wf id="w3" sent="1" para="1" offset="14" length="5">sells</wf>
    <wf id="w4" sent="1" para="1" offset="20" length="2">10</wf>
    <wf id="w5" sent="1" para="1" offset="22" length="1">%</wf>
    <wf id="w6" sent="1" para="1" offset="24" length="5">stake</wf>
    <wf id="w7" sent="1" para="1" offset="30" length="2">in</wf>
    <wf id="w8" sent="1" para="1" offset="33" length="7">Porsche</wf>
    <wf id="w9" sent="1" para="1" offset="41" length="2">to</wf>
    <wf id="w10" sent="1" para="1" offset="44" length="8">founding</wf>
    <wf id="w11" sent="1" para="1" offset="53" length="8">families</wf>
    <wf id="w12" sent="1" para="1" offset="61" length="1">.</wf>
  </text>
  <terms>
    <!--Qatar-->
    <term id="t1" type="close" lemma="Qatar" pos="R" morphofeat="NNP">
      <span>
        <target id="w1" />
      </span>
      <externalReferences>
        <externalRef resource="WordNet-3.0" reference="ili-30-08986905-n" confidence="1.0" />
        <externalRef resource="WordNet-3.0" reference="ili-30-08986691-n" />
      </externalReferences>
    </term>
    <!--Holding-->
    <term id="t2" type="close" lemma="Holding" pos="R" morphofeat="NNP">
      <span>
        <target id="w2" />
      </span>
      <externalReferences>
        <externalRef resource="WordNet-3.0" reference="ili-30-00810598-n" confidence="1.0" />
        <externalRef resource="WordNet-3.0" reference="ili-30-13244109-n" />
      </externalReferences>
    </term>
  </terms>
</NAF>
```


NAF: layered “stand off” Annotation

Qatar Holding sells 10% stake in Porsche to founding families

```
<entity id="e1" type="ORGANIZATION">
  <references>
    <!--Qatar Holding-->
    <span>
      <target id="t1" />
      <target id="t2" />
    </span>
  </references>
  <externalReferences>
    <externalRef resource="spotlight_v1" reference="http://dbpedia.org/resource/Qatar_Investment_Authority" confidence="1.0" reftype="en" />
  </externalReferences>
</entity>
<entity id="e2" type="ORGANIZATION">
  <references>
    <!--Porsche-->
    <span>
      <target id="t8" />
    </span>
  </references>
  <externalReferences>
    <externalRef resource="spotlight_v1" reference="http://dbpedia.org/resource/Porsche" confidence="0.9962352" reftype="en" />
    <externalRef resource="spotlight_v1" reference="http://dbpedia.org/resource/Porsche_family" confidence="0.0037167938" reftype="en" />
    <externalRef resource="spotlight_v1" reference="http://dbpedia.org/resource/Porsche_911" confidence="1.6859167E-5" reftype="en" />
    <externalRef resource="spotlight_v1" reference="http://dbpedia.org/resource/Ferdinand_Porsche" confidence="1.6186628E-5" reftype="en" />
    <externalRef resource="spotlight_v1" reference="http://dbpedia.org/resource/Porsche_914" confidence="7.881214E-6" reftype="en" />
    <externalRef resource="spotlight_v1" reference="http://dbpedia.org/resource/Porsche_Design_Group" confidence="7.0771493E-6" reftype="en" />
    <externalRef resource="spotlight_v1" reference="http://dbpedia.org/resource/Porsche_in_motorsport" confidence="2.7061944E-9" reftype="en" />
    <externalRef resource="spotlight_v1" reference="http://dbpedia.org/resource/Porsche_550" confidence="7.388618E-12" reftype="en" />
    <externalRef resource="spotlight_v1" reference="http://dbpedia.org/resource/Porsche_3512" confidence="1.6147437E-12" reftype="en" />
    <externalRef resource="spotlight_v1" reference="http://dbpedia.org/resource/Porsche_RS_Spyder" confidence="1.5382413E-12" reftype="en" />
  </externalReferences>
</entity>
....
```

Qatar Holding is interpreted according to this annotation as entity of the type ORGANIZATION, and it is linked to dbpedia article: Qatar Investment Authority.

Creating the training data

Data annotation procedure

1. **Sampling**: Collect texts: e.g., tweets, news, blogs, books
2. Define an **annotation scheme** or **code book**:
 - tag set (e.g. PoS labels, emotions, entity types)
 - the unit of the annotation: word (or token), phrase, sentence, paragraph, document
 - criteria when to assign a tag to a piece of text
3. Train **human annotators** to use the annotation scheme or create a **crowd task**
4. Provide an **annotation tool** that loads texts and allow the annotator to assign tags
5. **Store** the annotations with the text, e.g., in XML or TAB separated

Quality of the annotations

- More than one person annotates the same data; in the case of crowd annotation 10 to 20 crowd workers
- Determine the Inter-Annotator-Agreement (IAA) by analysing multiple annotations (degree to which the annotators assigned the same label, corrected for the chance to assign the same label, depends on the tag set)
- Fix disagreements (adjudication): if IAA is too low (< 50 Kappa) the task is considered too difficult
- IAA is considered the upper ceiling of NLP (some exceptions: Native Language Identification);
- Can machines do better than humans?

BRAT: Annotation environment

[http://brat.nlplab.org/
examples.html#corpus-
examples-brat](http://brat.nlplab.org/examples.html#corpus-examples-brat)

Stanford CoreNLP

Output format: Visualise

Please enter your text here:

Chase Manhattan and its merger partner J.P.Morgan and Citibank, which was involved in moving about \$100 million for Raul Salinas de Gortari, brother of a former Mexican president, to banks in Switzerland, are also expected to sign on.

Submit Clear

Part-of-Speech:

1 Chase Manhattan and its merger partner J.P.Morgan and Citibank, which was involved in moving about \$100 million for Raul Salinas de Gortari, brother of a former Mexican president, to banks in Switzerland, are also expected to sign on.

NNP NNP CC PRP\$ NN NN NNP NNP CC NNP , WDT VBD VBN IN VBG RB
DOLLAR CD CD IN NNP NNP IN NNP , NN IN DT JJ JJ NN , TO NNS IN
\$ 100 million for Raul Salinas de Gortari, brother of a former Mexican president, to banks in
NNP , VBP RB VBN TO VB IN
Switzerland, are also expected to sign on.

Named Entity Recognition:

1 Chase Manhattan and its merger partner J.P.Morgan and Citibank, which was involved in moving about \$100 million for Raul Salinas de Gortari, brother of a former Mexican president, to banks in Switzerland, are also expected to sign on.

Organization Organization Org MONEY
Person Location Location
for Raul Salinas de Gortari, brother of a former Mexican president, to banks in Switzerland, are also expected to sign on.

Coreference:

1

Chase Manhattan and its merger partner J.P.Morgan and Citibank, which was involved in moving about \$100 million for Raul are also expected to sign on.

Mention Mention Mention Mention

Basic dependencies:

1 Chase Manhattan and its merger partner J.P. Morgan and Citibank, which was involved in

dep cc poss conj conj rcmod nsubjpass auxpass prep pcomp
nn nn cc PRP\$ NN nn NN NNP nn NNP cc CC NNP , WDT VBD VBN prep IN
nsubjpass prep pobj

Gold, silver & bronze data

- **Gold data:** labels assigned by human annotators
- **Silver data:** labels proposed by a machine and validated by people
- **Bronze data:** labels generated by a machine and not validated:
 - 1 likes like V 2 root
 - 9 New New NNP 10 nn
 - 10 York York NNP 8 obj

	Bronze	Gold/Silver
1 likes like V 2 root		0
9 New New NNP 10 nn		B-Location
10 York York NNP 8 obj		I-Location

 - used as training labels or as features in addition to the words (e.g., lemmas, PoS tags, dependencies annotations in addition to, e.g.,s IOB gold annotations)
 - **IMPORTANT:** train and test data need to get the same feature representations so also similar bronze data as features!!!!

What can be features of a text?

Anything (words, sequence of the words, POS, and many more), except for the (*gold*) label to be predicted because that would make the prediction circular

What can be features of a text?

- Text length, average sentence length, word length ...
- Case (word shape), punctuation, ...
- The (surrounding) words....
- Lexical word properties: part of speech, meaning, sentiment
- Sentence properties: syntax, semantic roles
- Document properties: topic, genre, publication date
- ***Any feature needs to be mapped to some vector representation!***

How to represent a text? Two ways:

- *Today, the president rejected new measures against the pandemic.*
- *On Monday, WHO declared the end of the pandemic and the measures.*

(two sentences with similar structure but different meaning)

- 1). As a **sequence** of tokens with a grammatical-semantic structure:

- [Adv]_{ADJ-TMP}, [Det Ns]_{NP-SUBJ}, [Vp [Adj Np [Prep [Det Ns]_{NP}]_{PP-MOD}]_{NP-OBJ}]_{VP}
- [Prep N]_{PP-TMP}, [Name]_{NP-SUBJ}, [Vp [Det Np [Prep [Det Ns Co Det Np]_{PP-MOD}]_{NP-OBJ}]_{VP}

ORDER

- 2). As a **bag** of down-cased lemmatised words:

- [against, new, measure, pandemic, president, reject, the, today]
- [and, declare, end, measure, monday, of, on, pandemic, the, who]

**NO
ORDER**

- As a bag of lemmatised words without stopwords:

- [measure, pandemic, president, reject]
- [declare, measure, pandemic, WHO]

**NO
ORDER**

Words as features

Bag of words (BoW) model of text

No word order, words and their frequencies are fingerprints of what a text represents

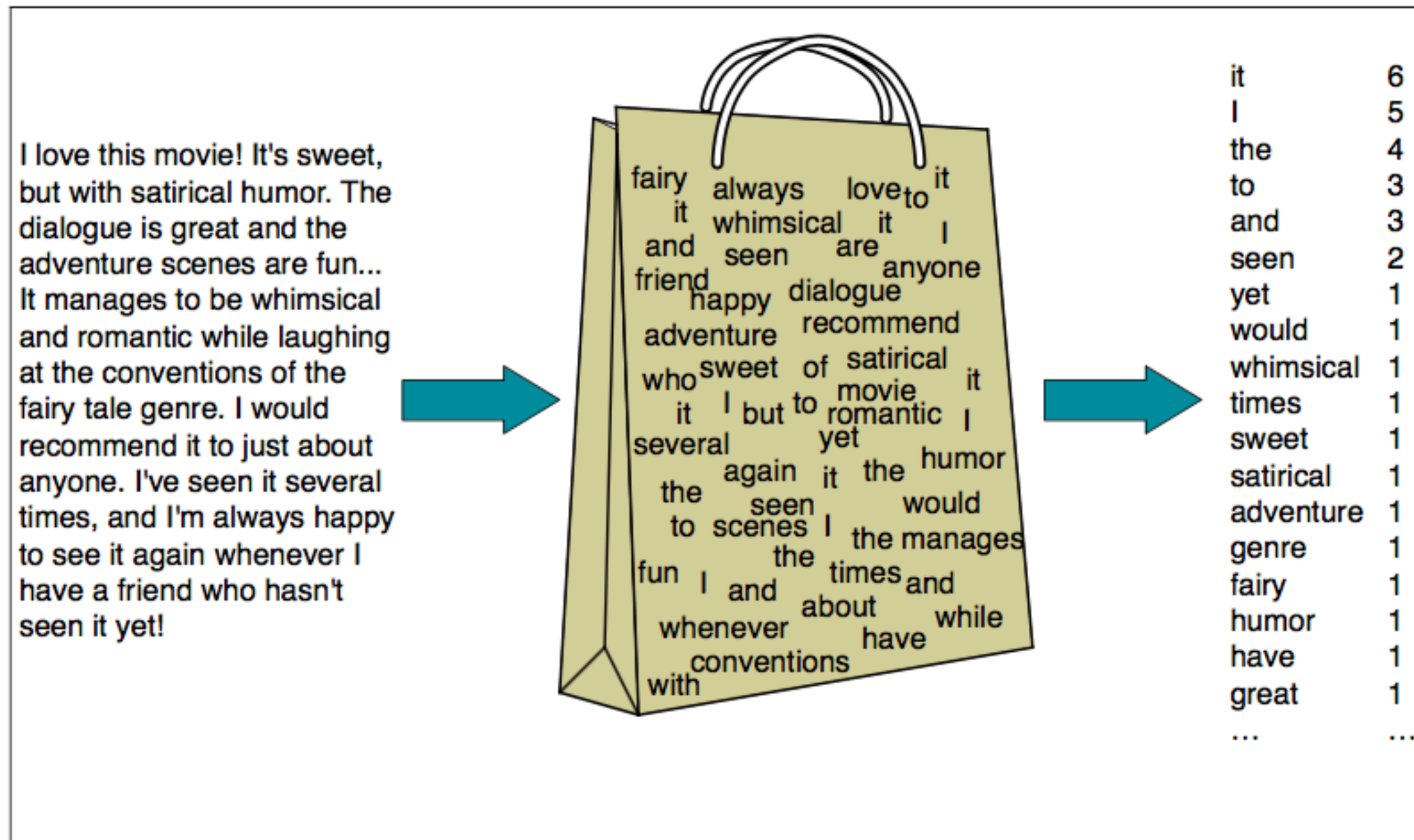


Figure 6.1 Intuition of the multinomial naive Bayes classifier applied to a movie review. The position of the words is ignored (the *bag of words* assumption) and we make use of the frequency of each word.

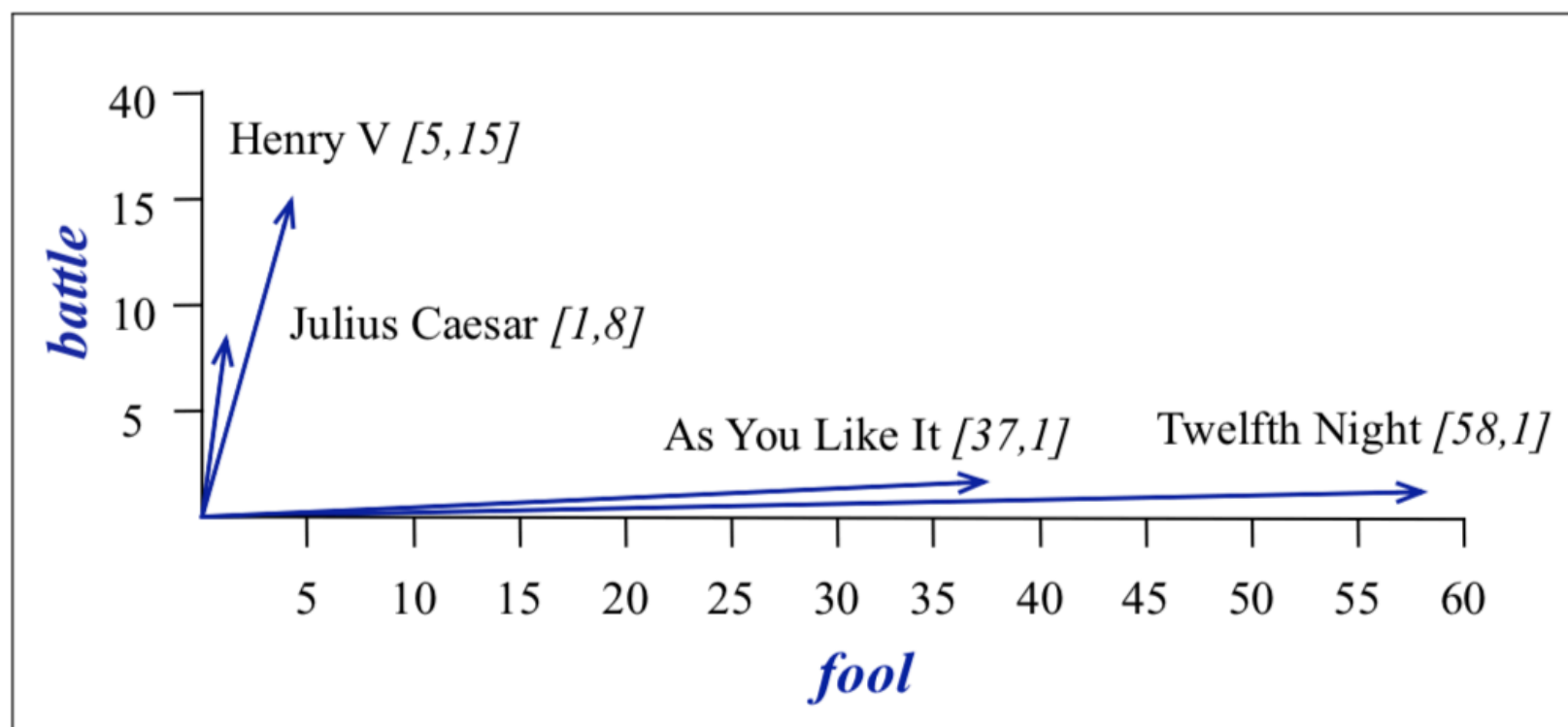
Word to Document index

(Jurafsky and Martin; vector semantics)

word vector representation of plays

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	1	8	15
soldier	2	2	12	36
fool	37	58	1	5
clown	5	117	0	0

Figure 15.2 The term-document matrix for four words in four Shakespeare plays. The red boxes show that each document is represented as a column vector of length four.



A text, such as a play, is represented as a vector in a multi-dimensional space

Figure 15.3 A spatial visualization of the document vectors for the four Shakespeare play documents, showing just two of the dimensions, corresponding to the words *battle* and *fool*. The comedies have high values for the *fool* dimension and low values for the *battle* dimension.

Not all words are equal

Information value (Spark-Jones 1972)

- Observations:
 - Words that occur in all texts, e.g. **a**, **the**, **case**, **of**, **is** have a low information value, despite their frequency
 - Words that occur once in one text are too idiosyncratic
 - Good words are typical for a topic or domain: shared to some extent but not too often
- $IV(t|d) = TF * IDF = \text{Term Frequency} * \text{Inverse Document Frequency}$

$$IV(t|d) = \frac{\text{Frequency of term } t \text{ in document } d}{\text{Total number of documents in which term } t \text{ occurs}}$$

- Powerful way to turn frequencies into information values

More than words and less than words

N-grams

- Sequences of 1, 2, 3, 4, 5 etc. words:
 - the, room, was, tidy (unigram, 1-gram)
 - the_room, room_was, was_tidy (bi-gram, 2-gram)
 - the_room_was, room_was_tidy (tri-gram, 3-gram)

The information could be richer, but more features and higher degree of data sparseness

- Character n-grams:
 - [th, he, e_, _r, ro, oo, om, m_, _w, wa, as, s_, _t, ti, id, dy]

Character 2-grams are more frequent than word 2-grams

- ***What is an advantage of character n-grams?***

Represent a bag-of-words

- Vector representations: a list or array with digital values

- Corpus:

- doc1: "Vaccines are safe"

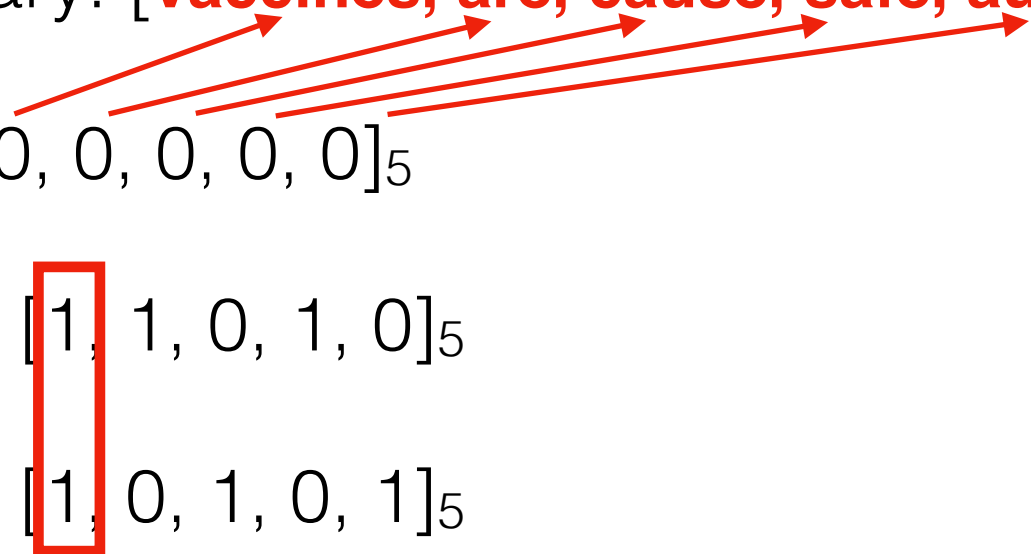
- doc2: "Vaccines cause autism"

- Vocabulary: [**vaccines, are, cause, safe, autism**]₅

- Vector: [0, 0, 0, 0, 0]₅

- doc1: [1, 1, 0, 1, 0]₅

- doc2: [1, 0, 1, 0, 1]₅



Bag-of-bi-grams

- Vector representations: a list or array with digital values
- Corpus:
 - doc1: “Vaccines are safe”
 - doc2: “Vaccines cause autism”
- Vocabulary: [**vaccines-are, are-safe, vaccines-cause, cause-autism**]₄
- Vector: [0, 0, 0, 0]₄
 - doc1: [1, 1, 0, 0]₄
 - doc2: [0, 0, 1, 1]₄

Bag-of-Words representation of a tweet with $TF*IDF$ values

[illegible]

Bag of words (n-gram) representations for text classification

- Represent a complete text (article, blog, tweet) as a feature bundle (as a set of features)
- Associate an interpretation label to the text as a whole, e.g. sentiment or the topic
- Learn the relation between a BoW representation and the label
- Predict the text label for the BoW representation of any unseen text (represent the unseen text using exactly the same vector representation, where we only have words from the training data, words that are not in the training data are ignored)
- Unsupervised: represent a text and measure text similarity, clustering (topic modelling) compare texts on the basis of feature overlap

One-hot encodings of words as vector values

Each token has an unique vector position, number of the dimensions == length of vocabulary

	autism	cause	is	safe	vaccination	vaccines
autism	= [1 , 0 , 0 , 0 , 0 , 0]					
cause	= [0 , 1 , 0 , 0 , 0 , 0]					
is	= [0 , 0 , 1 , 0 , 0 , 0]					
safe	= [0 , 0 , 0 , 1 , 0 , 0]					
vaccination	= [0 , 0 , 0 , 0 , 1 , 0]					
vaccines	= [0 , 0 , 0 , 0 , 0 , 1]					

Word vectors become large
and are sparse, even if
restricted to words with high
information value.

What about n-grams?

vector values:
● presence
● counts
● weights

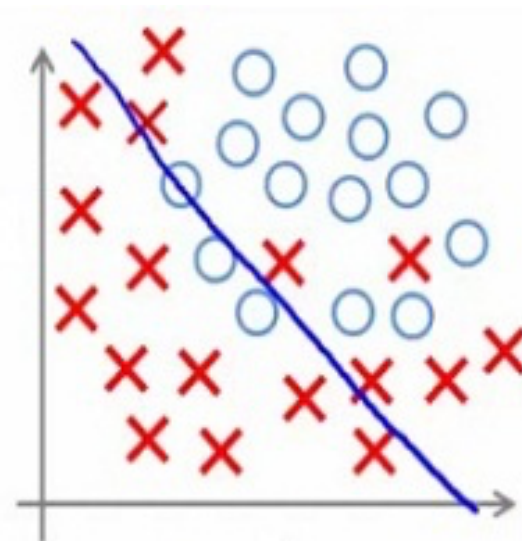
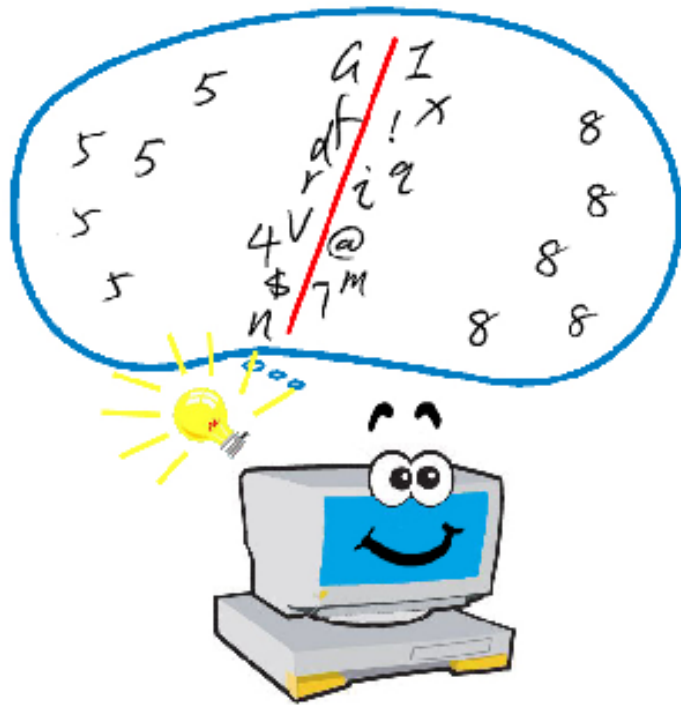
Encoding a complete text

“Vaccines cause autism” = [1 , 1 , 0 , 0 , 0 , 1] **Label = against**

“Vaccination is safe” = [0 , 0 , 1 , 1 , 1 , 0] **Label = pro**

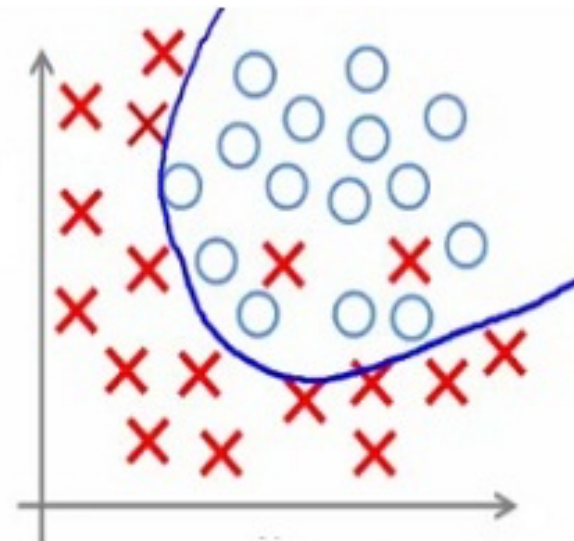
Dividing the feature space

Fitting the data to make predictions

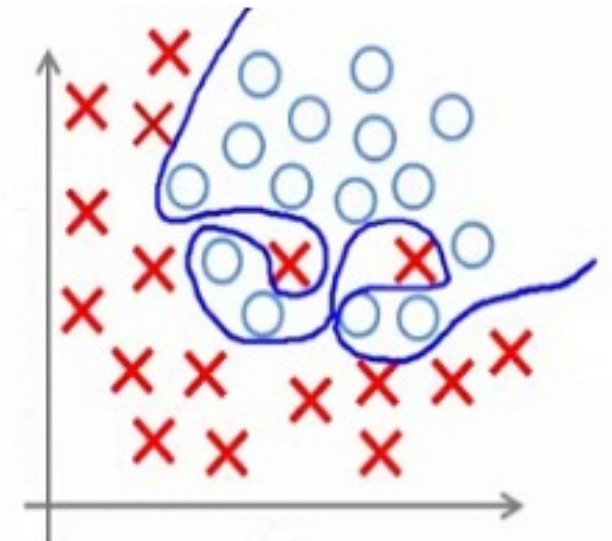


Under-fitting

(too simple to explain the variance)



Appropriate-fitting



Over-fitting

(forcefitting -- too good to be true)

Machine learning is about dividing the feature space that you defined

System needs to learn to draw the line between presence or absence of features in association with a particular class

Under-fitting: it cannot find enough properties to associate with the class (the model cannot explain why the presence of a particular feature is a good indication of a class)

Overfitting: learn about a feature that is very idiosyncratic for the training data

Types of fitting

- Probabilistic Classifier = assign the most probable class **c** to a text **t**
- **Generative classifiers:** e.g., Naive Bayes build a model for each class predicts which words are most likely given the class
- **Discriminative classifiers:** e.g., logistic regression learns how useful specific features are to predict a certain class
- More in Jurafsky and Martin!

Feature engineering

- Choose the properties of a text to become features or feature functions represented in a text vector
- Run (often many) experiments to choose the features that work best
 - feature ablation: define different sets of features and test what works best
 - bigrams or unigrams or lemmas + their part-of-speech or dependencies
 - represent multiword expressions (collocational, idiomatic), split compounds
 - frequency selections of words, stop word removal

	Unigrams	Bi-grams	Lemmas	UniPoS	Uni NoStopw	Uni Freq10	Uni Freq 100	Uni deps
Score	0,78	0,56	0,72	0,69	0,81	0,76	0,72	0,64

Machine learning

- Is about fitting the data, adjusting a model to avoid errors and make the right predictions
- (Some) problems:
 - Data sparseness (underestimated variation and dynamics)
 - Overfitting: you learn specifics from the training data that should not be learned (e.g., name of the file)

Disadvantage Supervised ML

- Requires handcrafted annotations (supervision)
 - time-consuming and expensive
 - again for each task/domain
 - again every 5 or 10 years due to language change and changes in the world
- Unsupervised learning can be used to create Language Models that solve these problems to some extent